Cryptanalyzing an Image-Scrambling Encryption Algorithm of Pixel Bits

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An efficient method can disclose important visual information concealed by a typical image-scrambling encryption algorithm in a ciphertext-only attack scenario. Experiments also show that attackers can use the real scrambling domain to support an efficient known- or chosen-plaintext attack.

Position scrambling (permutation) is one of the simplest and most efficient methods for protecting all kinds of multimedia data.[1,2] More importantly, it can effectively facilitate an encryption scheme to obtain an efficient combination of the properties of confusion and diffusion. Over the past two decades, researchers have proposed many multimedia scrambling algorithms, designing different mechanisms to derive the position-scrambling relation of the scrambling elements from a secret key. Because the number of possible scrambling relations is the factorial of the number of scrambling elements, there are numerous different scrambling relations in theory when the number of scrambling elements is sufficiently large. As for image, video, and audio (speech), various forms of datum can be selected as scrambling elements, including the bit, pixel, and compressing coefficients of an image or video,[3,4] or the transform coefficients and motion vectors of video[5] or audio frames.[6] If the scrambling elements are spatial blocks of an image, the recovery of the original image can be classified as part of the general problem of solving jigsaw puzzles.[7]

In contrast to cryptography, which is designing an algorithm to realize secure communication between the sender and intended recipient, the object of cryptanalysis is to gain as much information about the secret key as possible without any prior knowledge of it.[8–10] Cryptanalytic research into the proposed multimedia encryption schemes encourages designers to better understand cryptanalytic techniques. This would help them combine classic text-encryption schemes and multimedia properties for any given application scenario to meet required security levels. As a typical position permutation-only image cipher, Guodong Ye's proposed image-scrambling encryption algorithm (ISEA)[11] has received extensive attention from researchers working on the two basic parts of cryptology (cryptography and cryptanalysis).

Here, we re-evaluate the security of the simpler version of ISEA[11] and find the real reasons behind its attack advantage[12]: it does not act on a scrambling domain of size $M \times (8N)$, but simply cascades two scrambling algorithms with scrambling domains of sizes M and $8N$. We therefore propose an efficient known-plaintext attack and a general chosen-plaintext attack on the algorithm. We also found that important visual information of the plain image can be observed from a single cipher image encrypted by ISEA).

Cryptanalysis of Multimedia Scrambling Algorithms

Specific multimedia scrambling encryption algorithms have been successfully cryptanalyzed under different conditions in terms of the number of required plaintexts, the attack's computational complexity, and the storage space needed.[13–15] Because different scrambling elements can have dramatically different effects on plaintext's sensible information, different multimedia scrambling encryption algorithms might possess totally different strengths against ciphertext-only attacks. However, it's possible to evaluate their strengths against plaintext attacks using a uniform model.[3]

Any multimedia scrambling encryption algorithm can be efficiently broken with $O(\lceil \log_L MN \rceil)$ plaintexts and $O(\lceil \log_L MN \rceil \cdot MN^2)$ computational time, where MN is the

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size of the *scrambling domain* (that is, the position-scrambling scope of the scrambled elements), L is the number of different value levels of the scrambled elements, and $\lceil \cdot \rceil$ returns the smallest integer greater than or equal to a given number.[3] In other work, the computational load is further reduced to $O(\lceil \log_L(MN) \rceil \cdot MN)$ by replacing the operations of intersecting sets of position candidates with linear visits of a branch tree that stores information about the position candidates in its node.[16]

Guodong Ye and his colleagues proposed an ISEA that scrambles a binary presentation of a gray-scale plain image with a pseudorandom number sequence generated by iterating a digital chaotic map.[17] In 2011, ISEA was analyzed as a scrambling encryption algorithm acting on a scrambling domain of size $M \times (8N)$.[16] A weaker version of ISEA was suggested by Ye and his colleagues to decrease the computation complexity and save running time; in it, every row and column of the binary matrix of size $M \times (8N)$ are scrambled with the same scrambling vector.[17] In 2012, Liang Zhao and his colleagues built a set of specific plaintexts to recover the equivalent version of the secret key of ISEA.[18]

The ISEA Algorithm

ISEA's encryption object is a gray-scale image of size $M \times N$ (height $\times$ width), which can be represented as a matrix over the set $\{0, 1, \cdots, 255\}$: $\boldsymbol{I} = [I(i,j)]_{i=0,j=0}^{M-1,N-1}$. The image $\boldsymbol{I}$ is further decomposed as an $M \times (8N)$ binary matrix $\boldsymbol{B} = [B(i,l)]_{i=0,l=0}^{M-1,8N-1}$, where

$$\sum_{k=0}^{7} B(i,l) \cdot 2^k = I(i,j), \tag{1}$$

and $l = 8 \cdot j + k$. After preforming a scrambling of the row and column vectors of $\boldsymbol{B}$, the cipher image $\boldsymbol{I'} = [I'(i,j)]_{i=0,j=0}^{M-1,N-1}$ is obtained, where

$$I'(i,j) = \sum_{k=0}^{7} B'(i, 8 \cdot j + k) \cdot 2^k.$$

ISEA's basic concrete parts can be described as follows.

The Secret Key

First, the *secret key* is three positive integers m, n, and T_i. The initial condition is $x_0 \in (0,1)$, and the control parameter is $\mu \in (3.569945672, 4)$ of the Logistic map

$$f(x) = \mu \cdot x \cdot (1 - x). \tag{2}$$

Initialization

Next, initialization consists of three steps.

The first step is to run the Logistic map from x_0 to generate a chaotic sequence, $\{x_k\}_{k=1}^{L}$, where $L = \max\{(m + M), (n + 8N)\}$.

The second step is to produce a vector $\boldsymbol{T_M}$ of length M, where $\boldsymbol{S_M}(\boldsymbol{T_M}(i))$ is the $(i + 1)$-th largest element of $\boldsymbol{S_M} = \{x_{m+k}\}_{k=1}^{M}$, $i \in \{0,1, \cdots, M - 1\}$.

The final step is to produce a matrix $\boldsymbol{T_N}$ of size $1 \times (8N)$, where $\boldsymbol{S_N}(\boldsymbol{T_N}(j))$ is the $(j + 1)$-th largest element of $\boldsymbol{S_N} = \{x_{n+k}\}_{k=1}^{8N}$, $j \in \{0, 1, \cdots, 8 \cdot N - 1\}$.

Encryption

The encryption procedure also consists of three steps.

The first step is *vertical permutation*: generate an intermediate matrix $\boldsymbol{B}^* = [B^*(i,l)]_{i=0,l=0}^{M-1,8N-1}$, where

$$B^*(i,:) = B(\boldsymbol{T}_M(i),:). \tag{3}$$

The second step is *horizontal permutation*: generate an intermediate matrix $\boldsymbol{B'} = [B'(i,l)]_{i=0,l=0}^{M-1,8N-1}$, where

$$B'(:,l) = B^*(:,\boldsymbol{T}_N(l)). \tag{4}$$

The third step is *repetition*: reset the value of x_0 with the current state of the Logistic map (Equation 2) and repeat the above initialization operations and the two encryption steps $(T_i - 1)$ times.

Decryption

Finally, the decryption procedure is similar to the encryption procedures, with three simple modifications: the multiple rounds of encryption are executed in the reverse order; step 2 is executed first in each round; and we swap the left and right parts of Equations 3 and 4.

(a) (b)

FIGURE 1. A pair of plain and cipher images: (a) the cipher image and (b) the original "Lenna" image.

Cryptanalysis of ISEA

The full version of ISEA is given elsewhere,[16] where the matrix $\boldsymbol{T}_N$ is of size $M \times (8N)$ and each row vector $B^*(i,:)$ is scrambled by the i-th row vector of $\boldsymbol{T}_N$. Because this earlier work comprehensively cryptanalyzed the full ISEA version, we examine only the simpler version here.

Ciphertext-Only ISEA Attack

The ciphertext-only attack is an attack model for cryptanalysis in which the attacker can access only some ciphertexts. Obviously, possessing sufficient robustness to withstand the attack is a basic requirement for any multimedia encryption algorithm. Unfortunately, ISEA fails to satisfy that requirement.

ISEA's essential form of encryption can be represented by

$$\boldsymbol{B}' = (\boldsymbol{T}_L)^{T_i} \cdot \boldsymbol{B} \cdot (\boldsymbol{T}_R)^{T_i},$$

where $\boldsymbol{T}_L$ and $\boldsymbol{T}_R$ are permutation matrices representing the vertical permutation in Equation 3 and the horizontal permutation in Equation 4, respectively.[12] As is well known, the multiplication product of any number of arbitrary permutation matrices is still a permutation matrix—that is, a square binary matrix possessing exactly one entry of 1s in each row and each column. Regardless of the value of $\boldsymbol{T}_i$ and whether the intermediate matrices $\boldsymbol{B}^*$ in Equation 3 and $\boldsymbol{B}'$ in Equation 4 are updated in each round, the final essential form of ISEA's encryption process is still

$$\boldsymbol{B}' = (\hat{\boldsymbol{T}}_L)^{T_i} \cdot \boldsymbol{B} \cdot (\hat{\boldsymbol{T}}_R)^{T_i}, \quad (5)$$

where $\hat{\boldsymbol{T}}_L$ and $\hat{\boldsymbol{T}}_R$ are permutation matrices. So, repeating the operations in the third step of encryption has no influence on obtaining the equivalent version of ISEA's secret key. Without loss of generality, we just set the number of repetitions $T_i = 1$ in the following cryptanalysis.

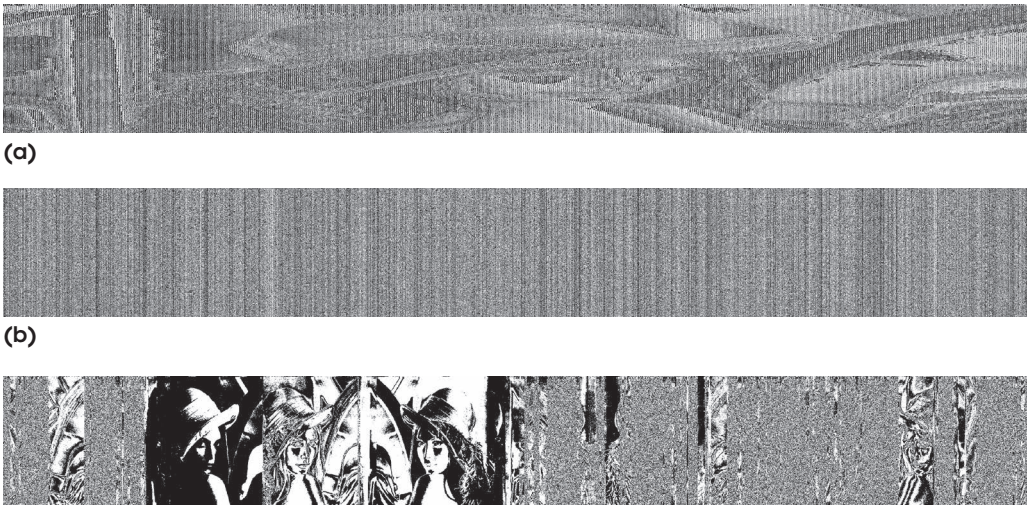
To facilitate the description, we use the cipher-image shown in Figure 1a as the attack object. The decomposition result of the image shown in Figure 1b using Equation 1 is shown in Figure 2a. Comparing Figures 2a and 2b shows that most of Figure 2a's visual information is concealed by the scrambling operations. However, the correlations between the image's rows and columns are unchanged. Given a row vector or a column vector of the cipher image, its neighboring vector can be found by searching for the vector possessing the highest correlation index. Repeating this process iteratively in the horizontal and vertical directions gives us an approximate version of $\boldsymbol{B}$. Actually, we can view the vector search problem as a binary template matching problem.

For simplicity, we selected the ratio of the same bits existing in two binary vectors—that is, Sokal and Michener's measure[17]—as the similarity measure. The image in Figure 2c is the result obtained from Figure 2b using this approach. Some visible image blocks can be automatically detected with the image quality metric defined elsewhere,[18] where the image composed of the eight bit-planes of Figure 1a's image in the corresponding weight order is used as the reference.

Figure 3 shows four cropped binary images that reveal important visual information about the original image in Figure 1b (this is especially true of the rough sketch). As the third image in Figure 3 shows, the obtained result's vector order might reverse that of the original version because the attack method can confirm only the relative locations among the vectors.

Known-Plaintext Attack on ISEA

The known-plaintext attack is a cryptanalysis attack model in which the attacker knows every implementation detail of the analyzed encryption algorithm (Shannon's axiom) and can access a set of plaintexts and the corresponding ciphertexts encrypted with the same unknown

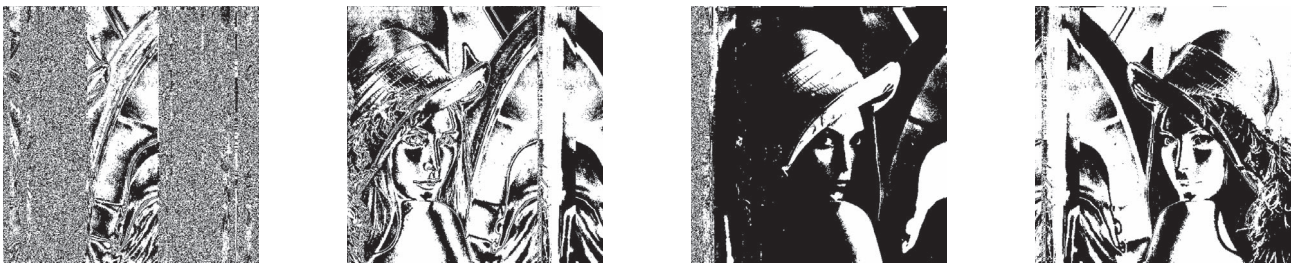


(a)

(b)

(c)

FIGURE 2. Intermediate images for breaking ISEA. (a) A binary presentation of the Lenna image in Figure 1b. (b) A binary presentation of the Lenna cipher image in Figure 1a. (c) The recovered binary image from Figure 2b.



(a) (b) (c) (d)

FIGURE 3. Four visible segments cropped from the image shown in Figure 2c. The obtained result's vector order might reverse that of the original version because the attack method can confirm only the relative locations among the vectors.

secret key. Because ISEA's two basic scrambling parts are exerted in two orthogonal directions, its ability to withstand the known-plaintext attack is weaker than other encryption algorithms that use only position permutation, with computational complexity of the same magnitude.

The equivalent version of the secret key of ISEA can be recovered using the traditional method, which involves the following:

- For each plain bit, put the positions of the cipher bits of the same value into the set containing its permutation position candidates.
- If more than one set of plain images and the corresponding cipher images are available, intersect the sets containing permutation-position candidates and obtain a final set for each plain bit.
- Assign the estimated permutation position of each plain bit with the first available (unassigned) element in its permutation position.

As noted elsewhere,[16] $O(\lceil \log_2(8MN) \rceil)$ known plain images can support efficient breaking of ISEA by considering it as a position-scrambling encryption algorithm exerting on a scrambling domain of size $M \times (8N)$ with permuted elements of two possible values. Essentially, ISEA is composed by cascading two basic permutation-only schemes with the

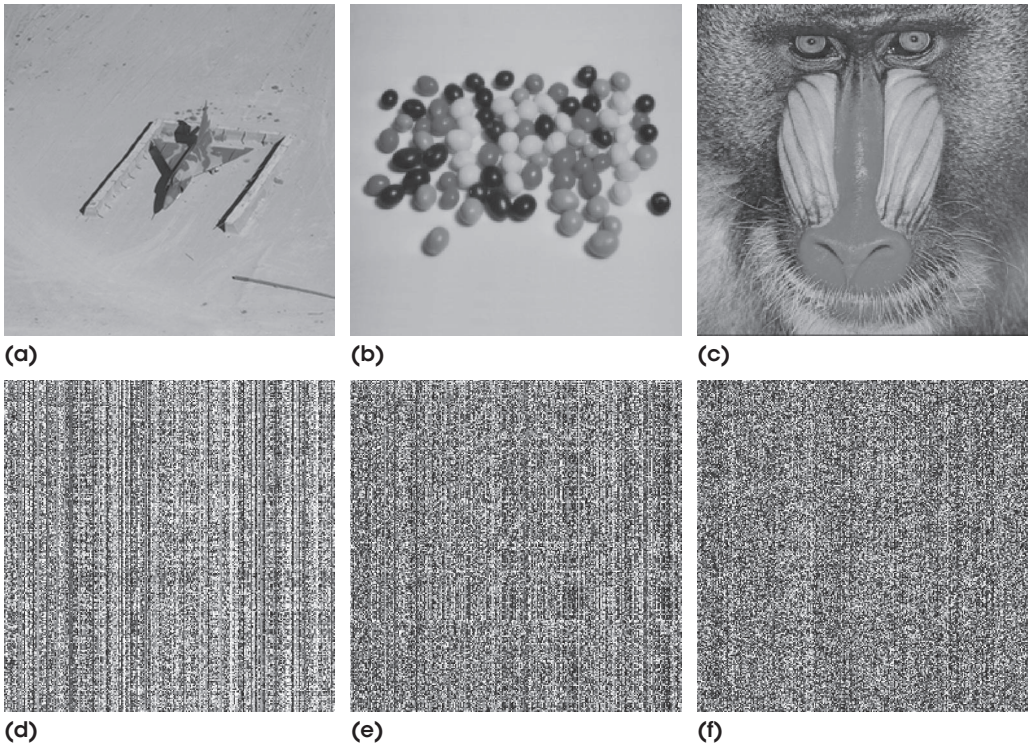


FIGURE 4. Three pairs of plain image and the corresponding cipher images: (a) "Fighter Jet," (b) "Candy," and (c) "Baboon," and (d) encrypted "Fighter Jet," (e) encrypted "Candy," and (f) encrypted "Baboon."

following parameters: a scrambling domain of size $M \times 1$, with permuted elements of $2^{(8N)} = 256^N$ possible values, and a scrambling domain of size $1 \times (8N)$, with permuted elements of 2^M possible values. Due to the cascading, the two basic permutation-only schemes cannot be broken separately in a direct way. However, some (and even all) parts of the schemes can be recovered iteratively. Concrete approaches involve the following five steps.

Step 1 involves comparing the number of "1" elements in each row vector of $\boldsymbol{B}'$ with that in all row vectors of $\boldsymbol{B}$. If $\boldsymbol{B}(j^*,:)$ is the unique row of $\boldsymbol{B}$ possessing the same number of 1 elements as $\boldsymbol{B}'(i^*,:)$, we can confirm $\boldsymbol{T}_M(j^*) = i^*$ and put i^* into set $\boldsymbol{R}$.

Step 2, similar to step 1, involves comparing the number of 1 elements in each column vector of $\boldsymbol{B}'$ with that in all $\mathbf{B}$ column vectors. If the former is unique in the latter, the corresponding element in $\boldsymbol{T}_N$ can be recovered correctly and the column number put into set $\boldsymbol{C}$ (the initial states of sets $\boldsymbol{R}$ and $\boldsymbol{C}$ are both empty).

Step 3 involves comparing each column vector of $\{\boldsymbol{B}^*(i,:)\}_{i=i_1^*}^{i_r^*}$ with all column vectors of $\{\boldsymbol{B}'(i,:)\}_{i=i_1^*}^{i_r^*}$ and adding its column number to set $\boldsymbol{C}$ if they contain the same unique column vector, where $\{\boldsymbol{B}^*(i,:)\}_{i=i_1^*}^{i_r^*}$ is recovered via Equation 3 and $\boldsymbol{R} = \{i_1^*, \cdots, i_r^*\}$.

Step 4, similar to step 3, involves comparing each row vector of $\{\boldsymbol{B}^*(:,j)\}_{j=j_1^*}^{j_c^*}$ with all row vectors of $\{\boldsymbol{B}(:,j)\}_{j=j_1^*}^{j_c^*}$ and adding its row number to set $\boldsymbol{R}$ if they contain the same unique row vector, where $\{\boldsymbol{B}^*(:,j)\}_{j=j_1^*}^{j_c^*}$ is obtained via Equation 4, and $\boldsymbol{C} = \{j_1^*, \cdots, j_c^*\}$.

Step 5 involves iteratively repeating steps 3 and 4 until the sizes of sets $\boldsymbol{R}$ and $\boldsymbol{C}$ both stop increasing (only newly found numbers are added into $\boldsymbol{R}$ and $\boldsymbol{C}$ in the above steps).

If more pairs of known plain images and the corresponding cipher images encrypted with the same secret key are available, the sets $\boldsymbol{R}$ and $\boldsymbol{C}$ can be further expanded with the above steps. The values of $\boldsymbol{T}_M(i^{**})$ and $\boldsymbol{T}_N(j^{**})$ can be assigned with the index of the first

vector possessing the same measure (the number of 1 elements in a vector or the vector itself) during the above comparisons, where $i^{**} \in (\mathbb{Z}_M - \boldsymbol{R})$ and $j^{**} \in (\mathbb{Z}_{8N} - \boldsymbol{C})$. As demonstrated in earlier work,[3,16] more known plain images can dramatically reduce the number of vectors with the same comparison measure, and $\boldsymbol{T}_M(i^{**})$ and $\boldsymbol{T}_N(\boldsymbol{j}^{**})$ are more likely to be correctly guessed; they can also be checked further with the methods described earlier.

To verify the attack's real performance, we use three pairs of known plain images of size 256×256 (see Figure 4) and the corresponding cipher images encrypted with secret key $m = 20$, $n = 51$, $x_0 = 0.2009$, and $\mu = 3.98$. Using the known plain image in Figure 4a, the sizes of $\boldsymbol{R}$ and $\boldsymbol{C}$ can reach 118 and 18, respectively, after step 2. Note that $\boldsymbol{B}$'s size is 256×2048. After the first and second operations in step 4, $\boldsymbol{R}$'s size further increases to 192 and 256, respectively. In contrast, $\boldsymbol{C}$'s size increases only to 1,809, 1,913, and 1,921 after the first three operations, respectively, in step 3. Because step 3's fourth operation does not increase $\boldsymbol{C}$'s size, we used the second plain image (Figure 4b), which increases it to 2,002. $\boldsymbol{C}$'s maximal value was obtained with the third plain image (Figure 4c). Figure 5 shows the ratios between the sizes of $\boldsymbol{R}$ and $\boldsymbol{C}$ and their respective maximal values during the above attack processes; the figure confirms our proposed known-plaintext attack's perfect performance.

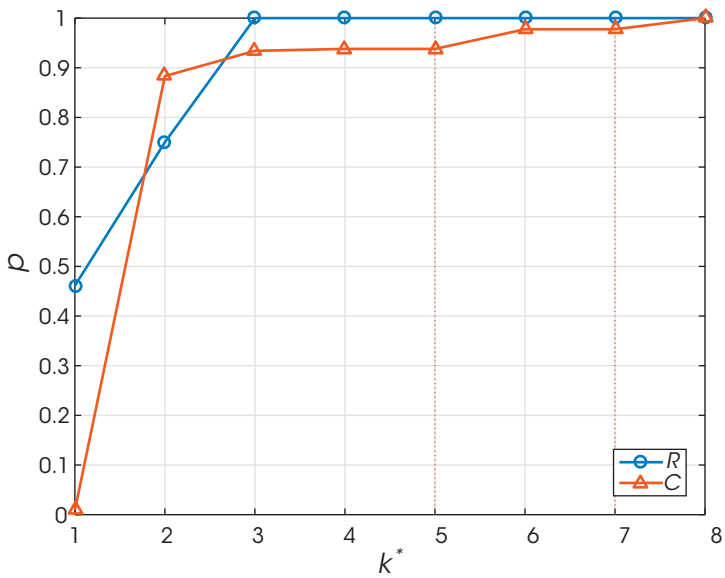


FIGURE 5. The ratios between the sizes of R and C and their maximal possible sizes during the attack with three pairs of plain images; k* denotes the index of the attacking steps.

Chosen-Plaintext Attack on ISEA

The chosen-plaintext attack is an enhanced version of the known-plaintext attack that lets attackers arbitrarily choose the plaintexts to optimize the breaking performance; they might, for example, reduce the number of needed plaintexts or increase the accuracy of the obtained information on the secret key. In work by Liang Zhao and his colleagues,[18] a set of specific plain images and the corresponding cipher images were selected to accurately obtain the scrambling relation in the two directions. However, we have found a way to further optimize this method, using some chosen plain images to break the scrambling operations in the two orthogonal directions at the same time.

If $M \leq 8N$, ISEA's equivalent secret key can be recovered with the following steps.

The first step is to choose a plain image satisfying $\mathbf{B} = [\boldsymbol{B}_L, \boldsymbol{B}_R]$, where

$$B_L = T_L \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \vdots \\ \vdots & \vdots & \ddots & 0 \\ 1 & \cdots & 1 & 1 \end{bmatrix}_{M \times M} \cdot T_R, \qquad (6)$$

$\boldsymbol{T}_L$ and $\boldsymbol{T}_R$ are any permutation matrices of size $M \times M$, and $\boldsymbol{B}_R$ is an $M \times (8N - M)$ matrix with constant elements zeroes or ones.

For step 2, because the number of ones in every row of $\boldsymbol{B}$ is different and the horizontal permutation has no influence on it, you can recover the vertical permutation vector $\boldsymbol{T}_M$ exactly by comparing the number of ones in each row of $\boldsymbol{B}$ to each row in $\boldsymbol{B}'$.

In step 3, ISEA has degenerated to a permutation-only scheme operating in a scrambling domain of size $8N$ with permuted elements of 2^M possible values. Referring to the fifth equation in earlier work,[1] we can ensure that the $8N$ elements of vector $\boldsymbol{T}_N$ can be exactly recovered with $\lceil \log_{2^M}(8N) \rceil$ chosen plain images, where $B_k(i,j) = \lfloor j/2^{M \cdot k+i} \rfloor \bmod 2$, $\lfloor x \rfloor$ gives the largest integer less than or equal

to x, and $k \in \{0,1,\cdots,\lceil \log_{2^M}(8N) \rceil - 1\}$, $i \in \{0, 1, \ldots, M-1\}$, and $j \in \{0, 1, \ldots, 8N-1\}$.

If $8N \in \{M, M+1\}$, we can use the plain image with $\boldsymbol{B}$ in Equation 6 to accurately confirm M elements of $\boldsymbol{T}_N$ with the same approach as in step 2, comparing the number of ones in each row. We can also recover the sole remaining element of $\boldsymbol{T}_N$ when $8N = M + 1$. So, the number of required chosen plain images for breaking ISEA is

$$\lceil \log_{2^M}(8N) \rceil = \left\lceil \frac{1}{M}(3 + \log_2 N) \right\rceil$$

if $M = 8N - 1$, and 1 if $8N \in \{M, M+1\}$. Similarly, we can deduce that the required number is

$$1 + \lceil \log_{2^{8N}} M \rceil = \left\lceil \frac{1}{8N}(\log_2 M) \right\rceil$$

if $M > 8N + 1$, and 1 if $M \in \{8N, 8N+1\}$. Ultimately, we can conclude that the number of required chosen plain images for breaking ISEA is

$$n^* = 1 + \begin{cases} 0 & \text{if } 8N \in \{M, M+1, M-1\}; \\ \left\lceil \frac{1}{M}(3 + \log_2 N) \right\rceil & \text{if } 8N > M + 1; \\ \left\lceil \frac{1}{8N}(\log_2 M) \right\rceil & \text{if } M > 8N + 1. \end{cases}$$

This is much smaller than the estimated number given in previous work,[12]

$$n' = \begin{cases} \lceil 8N / M \rceil + 1 & \text{if } M < N; \\ 9 & \text{if } M = N; \\ \leq 9 & \text{if } 8N \geq M > N; \\ \lceil M / 8N \rceil + 1 & \text{if } M > 8N. \end{cases}$$

Concretely, we have $n^* = 2$ if $8N + 1 < M \leq 2^{8N}$ or $M + 1 < 8N \leq 2^M$. As for an image of typical size 1704×2272, we can calculate

$$n^* = 1 + \left\lceil \frac{1}{1704}(3 + \log_2(2272)) \right\rceil = 2,$$

and $n' = \lceil 8 \cdot 2272/1704 \rceil + 1 = 11 + 1 = 12$. Because the computational load of the chosen-plaintext attack on a position permutation-only encryption algorithm is proportional to the product of the number of required plaintexts by their sizes, the attacking complexity of the proposed method is $O(n^* \cdot MN)$.[3]

As previous cryptanalytic works on the class of permutation-only encryption algorithms have shown, secret scrambling operations alone are incapable of providing sufficiently high security against known- and chosen-plaintext attacks.

Our cryptanalysis makes four additional contributions toward protecting multimedia data. First, the correlation in multimedia data can be used to support specific attacks and enhance breaking performance. Second, the size of each independent scrambling domain should be carefully checked to obtain the expected security requirement. Third, no matter what the permuted (scrambled) elements are, any permutation-only encryption algorithms will not change their histogram. Finally, the security level of a multimedia encryption algorithm should be suitable for a given application scenario. MM

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